

TMCS 96x32 Matrix — AI Sourcing Experiment (50 lines)

Input: **TMCS_96x32_Matrix_V1.2_test_raw.xlsx** — 25 lines with MPN, 25 without (seeded random). Ground truth: engineer-filled PartLink / LCSC PN in TMCS_96x32_Matrix_V1.2.xlsx. Date: 2026-07-05.

Arms

Arm A — Hybrid (production mirror): DigiKey/Mouser/element14 REST APIs pre-fetched by code (free), Haiku 4.5 judge via Batch API (50% off) + prompt caching; web search+fetch only for LCSC and API gaps.
Arm B — Browser agent: real Brave (CDP) loading live LCSC/DigiKey/Mouser/element14/Unikey pages from a residential IP; Haiku reads the pages and judges. Zero bot blocks encountered.

Cost & time

Arm A
₹123 total · ₹2.5/line
52 web searches · batch 50 lines

Arm B
₹38 total · ₹0.8/line
33 min wall clock · scraping free

100-line extrapolation
Arm A ≈ ₹247 · Arm B ≈ ₹77 + ~67 min

Scorecard vs engineer

	EXACT	FAMILY	AGENT+	DIFFERENT	MISS
Arm A (hybrid)	25	1	10	9	5
Arm B (browser)	25	1	11	10	3

EXACT = same LCSC PN or MPN as the engineer. FAMILY = variant of the engineer's part. AGENT+ = engineer left the line empty, agent found a valid option. DIFFERENT = disagrees with the engineer — needs human review (the agent may be better: check price/stock via the links). MISS = nothing usable.

Per-line results

Line
#3 C4
#4 C7,C8
#5 C9
#13 C20,C23
#18 C28,C29,C30
#20 C32
#23 C134
#24 C144
#25 C276,C277
#28 C293,C294
#30 D2,D56
#31 D3,D4
#33 D6
#40 D93
#42 F1,F2,F3,F4,F5,F6,F7,F8,F9,F10,F11,F12,F13,F14,F15,F16,F17,F18,F19,F20,F21,F22,F23,F24,F25,F26,F27,F28,F29,F30,F31,F32
#43 H1,H2,H3,H4,H5,H6,H7,H8,H9,H10,H11,H12,H13,H14,H15,H16,H17,H18,H19,H20,H21

#44 IC 1
#46 IC 10
#50 J16
#52 J18
#54 J20
#55 J21
#58 JP1,JP2
#60 L1
#62 L3
#64 Q1
#65 Q2,Q68
#67 Q9,Q10,Q11,Q12,Q13,Q14,Q21,Q22,Q23,Q24,Q25,Q26,Q33,Q34,Q35,Q36,Q37,Q38,Q45,Q46,Q47,Q48,Q49,Q50,Q57,Q58,Q59,Q60,Q61,Q62,Q65,Q66
#72 R4
#77 R10
#79 R12,R58,R59,R63,R67,R71,R72,R73,R74,R75,R76,R107,R108,R109,R110,R111,R112,R113,R114,R115,R116,R117,R118,R149,R150,R151,R152,R153,R154,R155,R156,R157,R158,R159,R160,R191,R192,R193,R194,R195,R196,R197,R198,R199,R200,R201,R202,R233,R234,R2:
#80 R13
#82 R15
#84 R18
#85 R19,R20,R38,R55,R68,R70,R295
#86 R21,R22,R23,R24,R25,R26,R27,R28,R29,R30,R31,R32,R33,R34,R35,R36,R39,R40,R41,R42,R43,R44,R45,R46,R47,R48,R49,R50,R51,R52,R53,R54,R56,R57,R60,R61,R83,R84,R85,R86,R87,R88,R89,R90,R91,R92,R93,R94,R125,R126,R127,R128,R129,R130,R131,R132,R133,R13:

#93 R96,R98,R100,R102,R104,R106,R138,R140,R142,R144,R146,R148,R180,R182,R184,R186,R188,R190,R222,R224,R226,R228,R230,R232,R264,R266,R268,R270,R272,R274,R290,R292
#96 R303,R310,R311,R312,R313,R314,R315,R316,R317,R318,R319,R320,R321,R322,R323,R324,R325,R326,R327,R328,R329,R330,R331,R332,R333,R334,R335,R336,R337,R338,R339,R340,R341
#97 R306,R307
#101 SW1
#102 SW2,SW3,SW4
#103 SW5
#104 U1,U2
#110 U40,U41,U42,U43,U44,U45,U64,U65,U66,U67,U68,U69,U88,U89,U90,U91,U92,U93,U112,U113,U114,U115,U116,U117,U136,U137,U138,U139,U140,U141,U160,U161
#111 U46,U47,U48,U49,U50,U51,U70,U71,U72,U73,U74,U75,U94,U95,U96,U97,U98,U99,U118,U119,U120,U121,U122,U123,U142,U143,U144,U145,U146,U147,U162,U163
#114 U168
#117 U171
#118 Y1
#120
#121

- element14's API rate-limited ~40% of Arm A pre-fetch calls (HTTP 403, free tier) — those lines leaned on DigiKey/Mouser/web.
- element14 prices are INR (India store); LCSC/DigiKey/Mouser USD. "Lowest cost" judged at $\text{INR} \approx \text{USD} \times 84$.
- Engineer links are digikey.in; agents used digikey.com — same parts, different storefront.
- DIFFERENT is not automatically wrong: several lines have many valid equivalents; the engineer picked one, the agent another. Spot-check via links.